MODULE 41 · EXAM STRATEGY - ALL DOMAINS

Practice exam 1

Twenty questions, weighted to the real domain split. Pause and answer before each reveal.

MODULE 41 of 43 DOMAIN Practice WEIGHT 20 questions

Before you start

- **Pause the video at each question.** The nine-second hold is a prompt, not the budget.
- **Give yourself two minutes per question**, which trains for the tighter beta pacing.
- Commit to an answer before unpausing. Write it down if that helps you be honest.
- Use the working order: read the ask, then the scenario, then the options.
- After each wrong answer, ask whether it was a **knowledge** error or a **reading** error.

> ON THE EXAM
>
> Module 39's distinction is the point of practising. Knowledge errors tell you what to revise; reading errors tell you what habit to change.

A logistics company receives vehicle telemetry from 4,000 vehicles, each emitting a record every ten seconds. The data must land in S3 in Parquet format for downstream training, with no custom processing applied in transit.

The team wants the least operational overhead and does not want to manage any consumer application.

Which ingestion approach should they use?

A Kinesis Data Streams with a custom consumer writing to S3

B Amazon MSK with a Kafka Connect S3 sink

C Kinesis Data Firehose delivering to S3 with format conversion

D Direct PUT to S3 from each vehicle, one object per record

◆ PAUSE AND DECIDE

Which ingestion approach should they use?

A Kinesis Data Streams with a custom consumer writing to S3

Streams suits custom processing, and the scenario says none is needed. A consumer application is exactly what they said they do not want to manage.

B Amazon MSK with a Kafka Connect S3 sink

Reasonable if they already ran Kafka. Nothing suggests they do, and it is considerably more to operate than Firehose.

C Kinesis Data Firehose delivering to S3 with format conversion

Firehose delivers to S3 with no consumer to manage, and converts to Parquet in transit. Both stated requirements, met directly.

D Direct PUT to S3 from each vehicle, one object per record

4,000 vehicles every ten seconds is 24,000 tiny objects a minute. The small-files problem from module 02, at scale.

THE SIGNAL

Two phrases decide it: 'no custom processing' and 'no consumer application to manage'. Both point away from Streams and MSK.

> An analytics team queries a 40 TB dataset of CSV files in S3 using Athena. Queries typically select six columns out of ninety. Costs are rising and queries are slow.
>
> The same data is also read by weekly training jobs.

What is the MOST cost-effective change?

A Convert the data to Parquet and partition it

B Move the data to S3 Glacier Instant Retrieval

C Load the data into Amazon Redshift

D Compress the CSV files with gzip

◆ PAUSE AND DECIDE

What is the MOST cost-effective change?

A Convert the data to Parquet and partition it

Athena bills by data scanned. Columnar storage reads six columns instead of ninety, and partitioning prunes further. It speeds the training reads too.

B Move the data to S3 Glacier Instant Retrieval

Reduces storage cost on data being queried regularly, and does nothing about scan volume, which is the actual driver.

C Load the data into Amazon Redshift

A warehouse to maintain and pay for continuously, to solve a problem a file format change solves. Not the most cost-effective.

D Compress the CSV files with gzip

Reduces scanned bytes somewhat and still reads all ninety columns. Gzip is also not splittable, which hurts parallel reads.

THE SIGNAL

'Six columns out of ninety' is the planted detail. It is the definition of a columnar-format problem.

A credit team trains a model using features that change over time, such as a customer's rolling 90-day spend. Their first model performed far better in validation than in production.

Investigation shows the training set used each customer's current feature values rather than the values as at the date of the historical decision.

What should they use to prevent this?

A Increase the size of the validation set

B Apply stratified sampling when splitting the data

C Use the Feature Store online store for training data

D Use a Feature Store point-in-time query against the offline store

◆ PAUSE AND DECIDE

What should they use to prevent this?

A Increase the size of the validation set

More data does not fix leakage. The validation set inherits the same future information the training set had.

B Apply stratified sampling when splitting the data

Addresses class balance across splits. It does nothing about a feature carrying information from after the decision.

C Use the Feature Store online store for training data

The online store holds only current values, which is precisely the problem described. It would guarantee the leakage rather than fix it.

D Use a Feature Store point-in-time query against the offline store

Point-in-time queries return each feature as it was at the given timestamp, so the training row matches what was actually known then.

THE SIGNAL

Better in validation than production, plus current values on historical rows, is time-travel leakage. It has one clean fix.

A fraud dataset has 0.4 percent positive cases. A data scientist oversamples the minority class in the full dataset, then splits it into training, validation and test sets.

The model reports excellent recall on the test set and performs poorly in production.

What went wrong?

A Oversampling should have been replaced by class weights

B Resampling was applied before the split, so duplicates crossed into the test set

C The positive class rate is too low for any model to learn from

D Recall is the wrong metric for an imbalanced problem

◆ PAUSE AND DECIDE

What went wrong?

A Oversampling should have been replaced by class weights

Class weights are a reasonable alternative and would not have caused this. The failure is the ordering, not the technique.

B Resampling was applied before the split, so duplicates crossed into the test set

Oversampling duplicates minority rows. Splitting afterwards puts copies of the same record in both training and test, so the test score measures memorisation.

C The positive class rate is too low for any model to learn from

0.4 percent is low and entirely workable. Fraud models routinely train at this balance.

D Recall is the wrong metric for an imbalanced problem

Recall is often exactly the right metric for fraud. The number is inflated here by contamination, not by being the wrong measure.

THE SIGNAL

Resample the training split only. Doing it before the split is one of the most common and most invisible errors in Domain 1.

A team is building a Bedrock knowledge base over 8,000 technical manuals containing tables, diagrams and long procedural sections. Answers must cite the source, and users frequently search using exact part numbers.

They have not yet created the data source.

Which TWO decisions should they get right before creating it?

A The chunking strategy, since it cannot be changed after the data source is connected

B The number of results returned, since it is fixed at creation

C The vector store, since hybrid search requires OpenSearch Serverless

D The generating model, since it cannot be changed later

◆ PAUSE AND DECIDE

Which TWO decisions should they get right before creating it?

A The chunking strategy, since it cannot be changed after the data source is connected

Correct. Chunking is fixed per data source, so changing it later means creating a new one and re-ingesting.

B The number of results returned, since it is fixed at creation

numberOfResults is a per-query parameter. It can be changed freely at any time.

C The vector store, since hybrid search requires OpenSearch Serverless

Correct. Exact part numbers need hybrid search, and hybrid requires an OpenSearch Serverless store with a filterable text field.

D The generating model, since it cannot be changed later

The generating model is chosen per request. Swapping it needs re-testing, not re-ingestion.

THE SIGNAL

The exact part numbers are the tell for C, and the irreversibility of chunking is the tell for A. Both are stated conditions.

> A healthcare organisation has received a new dataset from a partner for training. Before using it, they must establish whether it contains personal data, and quarantine it if so.
>
> The dataset is 900 GB across several thousand S3 objects in mixed formats.

Which service should they use?

A Amazon Macie

B SageMaker Clarify

C Bedrock Guardrails sensitive information filters

D AWS Glue DataBrew

◆ PAUSE AND DECIDE

Which service should they use?

A Amazon Macie

Macie discovers and classifies sensitive data at rest in S3, including personal data, and findings can drive automated action such as quarantining a bucket.

B SageMaker Clarify

Clarify measures bias and explains predictions. It does not scan storage for personal data, and it is closed to new customers.

C Bedrock Guardrails sensitive information filters

Guardrails evaluate prompts and responses in flight. They do not scan data sitting in S3.

D AWS Glue DataBrew

Profiles and transforms data interactively. It is not a sensitive-data discovery and classification service.

THE SIGNAL

In flight or at rest is the discriminator. Guardrails work on a request; Macie works on a bucket.

A model scores 0.97 on the training set and 0.71 on the validation set. The gap has widened over the last several training runs as the team added features.

The training data has 4,000 rows and the model now uses 180 features.

What is happening, and what should they try first?

A Underfitting. Add more features and train for longer

B Data drift. Retrain on more recent data

C Label noise. Re-label a sample of the training data

D Overfitting. Apply regularisation and reduce the feature count

◆ PAUSE AND DECIDE

What is happening, and what should they try first?

A Underfitting. Add more features and train for longer

Underfitting shows as poor scores on both sets. Training is at 0.97, and adding features is what widened the gap.

B Data drift. Retrain on more recent data

Drift is a production phenomenon comparing live data to a baseline. This gap is between training and validation, before deployment.

C Label noise. Re-label a sample of the training data

Noisy labels usually depress both scores. They would not produce 0.97 on training.

D Overfitting. Apply regularisation and reduce the feature count

High training score, much lower validation score, widening as features were added, on only 4,000 rows. Textbook overfitting from a high feature-to-row ratio.

THE SIGNAL

180 features on 4,000 rows is the planted detail. The ratio is the cause, and it is why reducing features is the first move.

A team runs hyperparameter tuning on a deep learning model. Each full training run takes about four hours, and their last tuning job of 60 trials cost more than their monthly budget allows.

Their training script emits a validation metric after every epoch.

Which tuning strategy should they use?

A Grid search, for reproducible coverage of the space

B Hyperband, which stops unpromising trials early

C Random search, for maximum parallelism

D Bayesian optimisation, which learns from previous trials

◆ PAUSE AND DECIDE

Which tuning strategy should they use?

A Grid search, for reproducible coverage of the space

Grid search is the most expensive strategy here: it runs every combination to completion regardless of how badly a trial is doing.

B Hyperband, which stops unpromising trials early

The script emits a per-epoch metric, which is exactly what Hyperband needs to kill a bad trial before it consumes four hours.

C Random search, for maximum parallelism

Parallelism reduces wall-clock time and not total compute. Sixty trials still run to completion, so the cost is unchanged.

D Bayesian optimisation, which learns from previous trials

Finds good values in fewer trials and runs each one to completion. Better than grid, and it does not use the per-epoch signal available here.

THE SIGNAL

'Emits a validation metric after every epoch' is the planted prerequisite. Hyperband cannot work without it, and it is stated.

A team is comparing two models that detect faulty components on a production line. Faults are 1.2 percent of items. A missed fault reaches a customer; a false alarm sends an item for a thirty-second manual check.

Inspection capacity comfortably exceeds the current flag rate.

Which metric should they optimise?

A Recall, because a missed fault is unrecoverable and inspection capacity is available

B Precision, to avoid wasting inspection time

C Accuracy, since both error types affect quality

D R squared, to show how much variance is explained

◆ PAUSE AND DECIDE

Which metric should they optimise?

A Recall, because a missed fault is unrecoverable and inspection capacity is available

A miss reaches a customer and nothing downstream catches it. A false alarm costs thirty seconds of capacity the scenario says is spare.

B Precision, to avoid wasting inspection time

Would be right if inspection were the bottleneck. The scenario states capacity comfortably exceeds the flag rate, which removes that objection.

C Accuracy, since both error types affect quality

At 1.2 percent positives, a model flagging nothing scores 98.8 percent. Accuracy cannot separate the two models.

D R squared, to show how much variance is explained

A regression metric. This is a binary classification problem.

THE SIGNAL

Two costs are described and one is stated as affordable. That asymmetry chooses the metric, exactly as module 17 said.

A team must choose between four foundation models for a service that drafts customer apology emails from a short case summary. Nobody has written reference emails, and producing enough of them would take weeks.

The business cares that drafts are accurate to the case, complete, and appropriately apologetic in tone.

How should they evaluate the four models?

A ROUGE against emails generated by the largest model

B BLEU, using previous agent emails as references

C LLM-as-a-judge on correctness, completeness and professional style and tone

D Perplexity, as a reference-free quality measure

◆ PAUSE AND DECIDE

How should they evaluate the four models?

A ROUGE against emails generated by the largest model

Treats one candidate's output as ground truth, so it measures similarity to that model and would rank it highest by construction.

B BLEU, using previous agent emails as references

There are many good ways to apologise. BLEU rewards matching one agent's exact words, which is not what quality means here.

C LLM-as-a-judge on correctness, completeness and professional style and tone

No reference needed, and those three metrics map directly onto the three things the business said it cared about.

D Perplexity, as a reference-free quality measure

Reference-free, and it measures fluency rather than quality. A fluent email that misses the case scores well.

THE SIGNAL

'Nobody has written reference emails' eliminates every classical metric at once. That one sentence does all the work.

An insurer must include, in every automated decline letter, the three factors that most influenced that specific decision and whether each raised or lowered the score.

The letter is generated at the moment of the decision. The model is XGBoost on a real-time endpoint.

What should they implement?

A. A partial dependence plot for each feature, included in the letter

B. Online explainability, returning per-instance SHAP values with the prediction

C. A nightly job computing global feature importance

D. A foundation model that writes an explanation of each decline

◆ PAUSE AND DECIDE

What should they implement?

A A partial dependence plot for each feature, included in the letter

A PDP is a global view of a feature's marginal effect. It says nothing about this applicant's decision.

B Online explainability, returning per-instance SHAP values with the prediction

Per-instance, real time, and SHAP values are signed so they give direction as well as magnitude. Both stated requirements.

C A nightly job computing global feature importance

Global scope and batch timing. The letter is about one applicant and is issued immediately.

D A foundation model that writes an explanation of each decline

Produces a plausible narrative rather than an attribution. A stated reason is not evidence of the computation.

THE SIGNAL

Two constraints are stated: per applicant, and at the moment of decision. Only one option satisfies both.

A genomics platform accepts sequence files averaging 300 MB. Analysis takes twelve to twenty minutes per file. Researchers upload files individually through the working day and are content to be notified when results are ready.

Volume is unpredictable, with some days seeing none at all.

Which inference option fits?

A Real-time inference on a large memory-optimised instance

B Serverless inference with 6 GB memory

C Batch transform, run every two hours

D Asynchronous inference, with results to S3 and SNS notification

◆ PAUSE AND DECIDE

Which inference option fits?

A Real-time inference on a large memory-optimised instance

300 MB exceeds the 25 MB payload limit and twenty minutes exceeds the 60-second timeout. It fails on both numbers.

B Serverless inference with 6 GB memory

4 MB and 60 seconds. It fails more comprehensively than real-time, and memory is not the constraint.

C Batch transform, run every two hours

Would work, and adds up to two hours before processing starts. Files arrive individually rather than as a dataset.

D Asynchronous inference, with results to S3 and SNS notification

Up to 1 GB and up to one hour covers both numbers, requests arrive individually, and it scales to zero on the quiet days.

THE SIGNAL

Three signals: payload over 25 MB, processing over 60 seconds, and requests arriving individually rather than as a dataset.

A marketing platform trains one propensity model per client and now has 1,800 of them, all XGBoost on the same container image. Most clients are scored a handful of times a day, unpredictably.

A one to two second delay on the first scoring of the day is acceptable. Cost is the stated priority.

How should they host these models?

A. A multi-model endpoint, with all artefacts under one S3 prefix

B. 1,800 serverless endpoints, one per client

C. 360 multi-container endpoints, five clients each

D. An inference pipeline chaining the models

◆ PAUSE AND DECIDE

How should they host these models?

A A multi-model endpoint, with all artefacts under one S3 prefix

Same framework, same image, thousands of models, sparse access, and a stated tolerance for first-call latency. Every condition matches.

B 1,800 serverless endpoints, one per client

Serverless removes idle cost per endpoint and leaves 1,800 endpoints to create, monitor and update.

C 360 multi-container endpoints, five clients each

Multi-container is for different frameworks. These share one, and 360 endpoints is still an enormous amount to manage.

D An inference pipeline chaining the models

Serial execution would send every request through all 1,800 models in turn. The models are independent.

THE SIGNAL

The sentence about tolerating one to two seconds is the permission slip. Without it, cold model loads would be the objection.

A team deploys an updated model using canary traffic shifting with a 15 percent canary and a 45-minute baking period. The new model returns HTTP 500 for roughly a quarter of requests because of a serialisation defect.

The deployment completes successfully. All traffic now reaches the broken model and the blue fleet has been terminated.

What went wrong?

A The canary was too small to surface the errors

B The baking period was too short for the defect to appear

C No CloudWatch alarms were configured, so nothing could trigger auto-rollback

D Canary mode does not support automatic rollback

◆ PAUSE AND DECIDE

What went wrong?

A The canary was too small to surface the errors

15 percent of traffic returning 500s on a quarter of requests is a very visible signal. Nothing was configured to notice it.

B The baking period was too short for the defect to appear

A quarter of requests failing shows within seconds. Forty-five minutes is ample, and a longer period changes nothing without an alarm.

C No CloudWatch alarms were configured, so nothing could trigger auto-rollback

Auto-rollback fires when a pre-specified alarm trips. With no alarms, nothing can trip, so the deployment proceeded exactly as instructed.

D Canary mode does not support automatic rollback

Factually wrong. All three traffic shifting modes support auto-rollback, and alarms are a prerequisite for all of them.

THE SIGNAL

'The deployment completes successfully' is the tell. That is what a guardrail with nothing to check looks like.

A support knowledge base retrieves twenty chunks per query. Analysis shows the passage that answers the question is in the retrieved set about 92 percent of the time, but often appears below position ten.

The generating model is given only the top five chunks.

What should they change?

A Raise numberOfResults from twenty to one hundred

B Enable reranking, and return the top five after reranking

C Apply metadata filters to narrow the candidate set

D Switch the embedding model to one with more dimensions

◆ PAUSE AND DECIDE

What should they change?

A. Raise numberOfResults from twenty to one hundred

Recall is already 92 percent, so retrieval is finding it. More candidates worsens the ordering problem and costs more tokens.

B. Enable reranking, and return the top five after reranking

Recall is good and ordering is bad. A cross-encoder scores the query and each passage together, which first-stage search cannot do.

C. Apply metadata filters to narrow the candidate set

Filtering restricts by attribute. Nothing suggests the wrong documents are retrieved, only that they are badly ordered.

D. Switch the embedding model to one with more dimensions

Would require re-embedding the whole corpus, and recall is already high. The weakness is ranking, not finding.

THE SIGNAL

92 percent recall with the answer below position ten is the textbook reranking case: finding works, ordering does not.

A nightly workflow extracts data with a Glue job, validates it with a Lambda function, calls a third-party enrichment API from an ECS task, then trains and registers a model, then publishes to SNS and updates DynamoDB.

Several branches depend on outcomes, and each branch needs different error handling. The team wants the ML steps to retain native lineage.

What should orchestrate this?

A SageMaker Pipelines, using LambdaStep for the non-ML services

B EventBridge rules chained together, one per stage

C A single Lambda function coordinating every step

D Step Functions, invoking a SageMaker pipeline for the ML portion

◆ PAUSE AND DECIDE

What should orchestrate this?

A SageMaker Pipelines, using LambdaStep for the non-ML services

Forces Glue, ECS, SNS and DynamoDB through LambdaStep wrappers. Pipelines is built for ML steps, not as a general workflow engine.

B EventBridge rules chained together, one per stage

EventBridge routes events well and expresses no branching, error handling or overall state. There would be no single view of the workflow.

C A single Lambda function coordinating every step

One function owning a multi-service nightly workflow, inside a 15-minute limit, with hand-rolled error handling.

D Step Functions, invoking a SageMaker pipeline for the ML portion

Native integration with Glue, Lambda, ECS, SNS and DynamoDB with per-branch error handling, and the ML part keeps its lineage.

THE SIGNAL

The combination is the answer: Step Functions for breadth, a SageMaker pipeline for the ML, each doing what it is built for.

A delivery time prediction model has become steadily less accurate over five months. Model Monitor reports no data quality violations, and every input feature distribution sits inside its baseline range.

The company opened three new distribution centres during that period and changed its courier partner.

What is happening?

A Data drift. The input distributions have shifted

B A feature pipeline fault producing incorrect values

C Concept drift. The relationship between features and delivery time has changed

D Model decay caused by an ageing container image

◆ PAUSE AND DECIDE

What is happening?

A Data drift. The input distributions have shifted

Data drift means the inputs moved. The scenario states every distribution sits inside its baseline range.

B A feature pipeline fault producing incorrect values

A fault would show as a data quality violation or a distribution shift. Monitor reports neither.

C Concept drift. The relationship between features and delivery time has changed

Inputs look normal and accuracy fell. New distribution centres and a new courier change what the same inputs imply about delivery time.

D Model decay caused by an ageing container image

A model artefact does not degrade with age. The image serves the same weights indefinitely.

THE SIGNAL

Normal inputs, falling accuracy, and a stated change in the world. That combination is concept drift every time.

A processing job that ran successfully for months is moved into a private subnet to satisfy a security standard. It now fails about seventy seconds after starting.

There are no log streams for the job in CloudWatch at all. The execution role and the S3 paths are unchanged.

What is the most likely cause?

A The subnet has no route to the services the job needs. Add VPC endpoints

B CloudWatch log delivery is delayed and the streams will appear shortly

C The execution role lost permissions when the VPC was attached

D Logging must be enabled per processing job and was not

◆ PAUSE AND DECIDE

What is the most likely cause?

A The subnet has no route to the services the job needs. Add VPC endpoints

A private subnet has no path to S3, ECR or CloudWatch without endpoints or NAT. The job fails before printing anything, which is exactly no logs.

B CloudWatch log delivery is delayed and the streams will appear shortly

Delivery is near real time. A seventy-second failure with no stream at all is not a delivery delay.

C The execution role lost permissions when the VPC was attached

Attaching a VPC does not modify IAM. The scenario states the role is unchanged.

D Logging must be enabled per processing job and was not

Anything a container writes to stdout or stderr goes to CloudWatch Logs automatically, with no configuration.

THE SIGNAL

No logs at all means the failure preceded the container printing anything. A missing network path is one of its four causes.

A research group runs GPU training jobs several times a week, with volume varying widely between quiet and busy weeks. Their scripts checkpoint every epoch. No individual job has a deadline.

Pipeline runs also repeat an unchanged 25-minute preprocessing step on every execution. Finance will not approve any term commitment.

Which TWO changes will reduce cost?

A Buy a one-year Savings Plan sized to average usage

B Enable Managed Spot Training, with max_wait above max_run

C Move training to a larger instance to finish faster

D Enable step caching on the pipeline

◆ PAUSE AND DECIDE

Which TWO changes will reduce cost?

A Buy a one-year Savings Plan sized to average usage

Finance has ruled out a commitment, and bursty volume makes an average-sized plan wasteful in quiet weeks anyway.

B Enable Managed Spot Training, with max_wait above max_run

Checkpointing exists, no job has a deadline, and volume is bursty. Every Spot condition is stated, for up to 90 percent off.

C Move training to a larger instance to finish faster

Raises the hourly rate to reduce duration. It rarely reduces total cost and is not what the scenario needs.

D Enable step caching on the pipeline

An unchanged 25-minute step reused from a previous run costs nothing. That is 25 minutes saved on every execution.

THE SIGNAL

Two separate cost problems are described, and each has its own answer. The commitment option is ruled out in the scenario text.

A pharmaceutical company will train a model on patient trial data using an algorithm container supplied by an external research partner. Their ethics board requires assurance that the partner's code cannot transmit any data outside the account.

The container image includes all its dependencies. Input and output are S3 paths in the company's own account.

What should they configure?

A VpcConfig, placing the job in a private subnet with S3 endpoints

B EnableNetworkIsolation on the training job

C EnableInterContainerTrafficEncryption on the training job

D A customer-managed KMS key on the training volume

◆ PAUSE AND DECIDE

What should they configure?

A VpcConfig, placing the job in a private subnet with S3 endpoints

A sensible layer, and the container can still call whatever those endpoints permit. It restricts the destination rather than forbidding calls.

B EnableNetworkIsolation on the training job

The container cannot make any outbound call, while SageMaker AI still stages input and output. Exactly the assurance the board asked for.

C EnableInterContainerTrafficEncryption on the training job

Encrypts traffic between instances in a distributed job. It does not stop a container calling out.

D A customer-managed KMS key on the training volume

Protects data at rest on the volume. It says nothing about what the container transmits.

THE SIGNAL

'Cannot transmit any data outside the account' is network isolation, not VPC mode. The dependencies line is what makes it viable.

After this set

- **16 or more out of 20** is a comfortable position. Work on the ones you missed.
- **12 to 15** is workable. Identify whether the misses cluster in one domain.
- **Under 12** means revisit the modules behind the wrong answers before the next set.
- **Classify every miss**: knowledge error, or reading error? They need different responses.
- The real exam is compensatory, so two weak domains cost more than one weak topic.

> ON THE EXAM
>
> This is not a predictor of your exam score. It is a diagnostic telling you where to spend the next few hours.